chapter I

This Building Does Not Exist –

An Attempt at a Theory of Neural Architecture

The people in this grid of faces do not exist. They are an example of the ability of Generative Adversarial Networks to create realistic images based on large datasets of faces.

This Building Does Not Exist - An Attempt at a Theory of Neural Architecture

What does a Melange have to do with Artificial Intelligence?

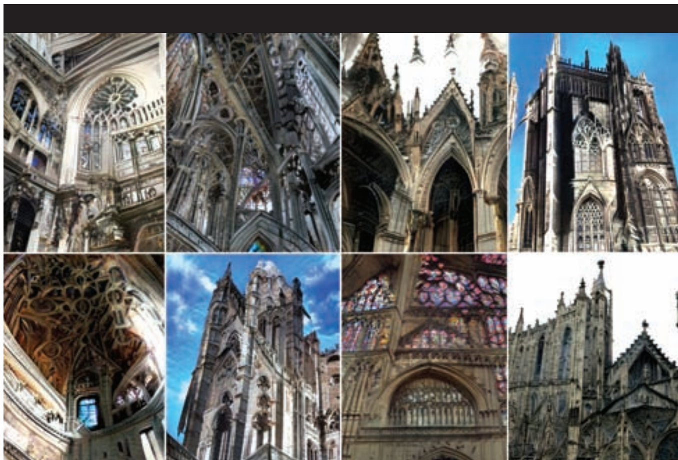
Vienna is home to one of the world's oldest clusters of research on artificial intelligence: the Austrian Institute of Artificial Intelligence (OFAI). The OFAI is an offspring of the Österreichische Studiengesellschafft für Kybernetik, a registered scientific society founded in 1969. About 30 years after its founding, in the summer of 1998, Sandra Manninger and I were sitting around in a Schanigarten, a typical Viennese coffee house guest garden, in the court of the Baroque ensemble that houses the OFAI in the heart of Vienna. We were discussing the possibilities of integrating artificial intelligence into an architectural design with Professor Robert Trappl, the director of the OFAI, and Dr. Arthur Flexer. Of course, this was a purely hypothetical conversation since neural networks based on computational processes were in their infancy—while sipping Melange coffees, they told us with excitement about a recent success: the simulation of one neuron to another neuron interaction. A few years later, in 2006, SPAN, in collaboration with the OFAI, conducted the first machine learning workshop at the Angewandte in Vienna. After a move to the University of Michigan in 2014, a collaboration with Michigan Robotics and Computer Science paved the way for a series of new design techniques. In particular, the director of Michigan Robotics, Jessy Grizzle, and PhD student Alexandra Carlson have been crucial to this collaboration.

Neural architecture is the field of architecture that is primarily preoccupied with interrogating the emergent field of Artificial Neural Networks (ANNs) as a method of designing architecture. ANNs can be described, in short, as a sequence of mathematical algorithms that are capable of registering latent correlations in a set of data. The applied algorithms mimic certain aspects of how the human brain operates. This position, however, is still highly disputed. For the sake of this book, I would suggest agreeing that the algorithms at work in Neural Networks (NNs) were inspired by the current knowledge about certain processes in the human brain, such as hallucinating[1] and dreaming.[2] In this sense, neural networks can be described as a system of neurons that can be either organic or artificial.

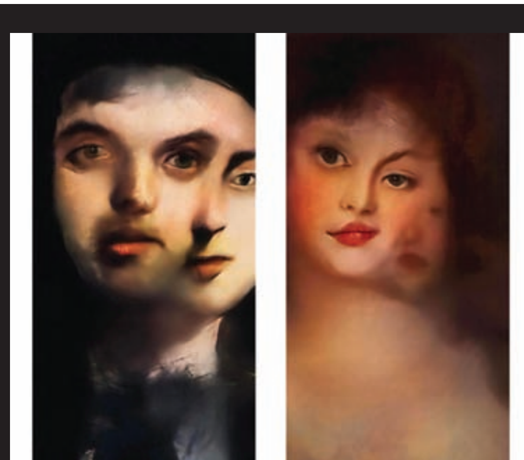
2015 was a turning point in the application of techniques derived from AI research to the arts. That year saw the introduction of the Generative Adversarial Network (GAN) by Ian Goodfellow,[3] as well as the publication of the paper *A Neural Algorithm of Artistic Style*, by Leon Gatys.[4] In recent years, these novel methods have taken hold in the arts[5] and music.[6] The newly emerging art form is fittingly named neural art[7]. The source of this term can be found in the title of a paper by Gatys,[8] which forms the base for the work of several of the most prolific neural artists, such as Mario Klingemann (who describes himself as a neurographer) and Sofia Crespo, whose series of works, *Neural Zoo*, reflects a keen interest in the estrangement and defamiliarization of deep-sea creatures. There are many more artists in this genre.[9]

How is the work of these artists related to architecture? Maybe an example will help clarify how provocative this novel method is for architecture. Mario Klingemann uses databases of Western art, particularly portraits, as the base for his StyleGAN[10] applications. Thousands of images from the Renaissance to the 19th century were fed through a StyleGAN algorithm. In a conventional case, StyleGAN would be used to create images that convincingly represent a known object to the observer. Most famously demonstrated with examples like *This Person Does Not Exist*, a website that generates highly convincing images of persons based on a large dataset of portraits. It might be important here to understand that neural networks are basically function approximation algorithms;[11] they will always strive to achieve an approximation of, for example, the number one. Function approximation can also be described by a curve, or as computer scientist and philosopher Judea Pearl pointed out: *"Machine Learning is just glorified 'Curve Fitting.'"*[12] As much as this approach can produce convincing images of objects, it is the area outside the perfect fit of the curve that really produces the more interesting results in that they maintain a certain familiarity, despite their alien appearance. Back to the work of Mario Klingemann: His images and animations maintain elements of the dataset informing the StyleGAN. This results in images that show contorted bodies and distorted faces that have a surreal quality to them—bizarre Janus heads with multiple faces, strange cyclops, and monstrous chimeras between humans and animals. The trained eye will still recognize features of historic paintings and drawings. A bit of Francisco Goya here, some James Whistler there, glimpses of Jeanne-Etienne Liotard, Eduard Magnus, Franz von Lenbach, and Franx Xaver Winterhalter.

Based on a large image dataset of Gothic churches, this GAN identified common features in the image set and interpolated them in order to create variations of the given image set. The adversarial network, in this case, was trained intentionally sloppy to avoid completely realistic images of Gothic churches, but rather to emphasize the possibility of an architecture that is strange but familiar at the same time.



As in the example of the Gothic churches, Mario Klingemann approaches the application of GAN's in his work *Memories of Passersby I* by avoiding the ability of GAN's to create realistic variations of classic portraits. He instead emphasizes the estrangement provoked by avoiding perfect curve fitting in the algorithm.



But none of the images are designed to approximate the particular artists, as *This Person Does Not Exist* would do. Rather, a StyleGAN renders the features that were recognized by the neural network and re-combines the pixels into a new image outside the conventions through which we, as humans, understand the depicted object. In doing so, the emergent pieces of art provoke questions of authorship and agency. In addition, the question is raised of the value of a sensibility that was created somewhere between human input and machinic output. Is this the art of the posthuman age? To illustrate an example in the field of architecture, we collected Gothic architecture images and used them as a dataset for a StyleGAN. Ironically, using the ever-so-popular scraping method to collect *"Goth"* images results in, let's say, *interesting* results. Does this development take into consideration the possibility of evaluating the role of humans in a world where the boundaries between human and non-human creativity are blurred? In architecture, we can observe a similar tendency, with architects increasingly picking up on novel techniques in machine learning and machine vision. I would describe this new tendency in architecture as neural architecture, borrowing from computer science and neuroscience as much as from the language used in the arts and music (neural art, neural music).

This dataset with several thousand images of Baroque plans forms the base for a process that reveals how a neural style transfer imprints the learned features from the Baroque plans to a modern plan.

Familiar but Strange: Bits, Pieces, Features & Neurons

I have relied on neural networks to train datasets in my own work, exploring them in a reverse flow of information, which means that algorithms that are normally trained to recognize objects, such as in automated cars, etc., are used to perform generatively instead of analytically. This is a similar technique to the ones used by the artists mentioned above. Unbeknownst to myself, for example, Sofia Crespo uses a very similar 2D to 3D style-transfer technique to the one I have been using in recent years.[13] To give you a more concrete example, allow me to rely on an image-based approach: using a database with several thousand images of Baroque plans, we experimented with how a neural style transfer would imprint the learned features from the baroque plans onto a modern plan. Echoing the conversation about the work of neural artists, we achieve results that are familiar, but strange at the same time. The nature of Baroque architecture reflected in the dynamic opposition and intersection of elliptical volumes of space, the thick pochés, undulating walls, and rich surface treatment, are features that are learned by the neural network and applied to modern plans. Despite the fact that this is a simple experiment, it was a proof of concept as to how this method does not simply imitate a style or create a crude collage of elements. Rather, it produces unexpected, highly provocative, and—yes—inspiring images.

The aesthetics of estrangement and defamiliarization form the bedrock of contemporary art, applying ML and machine vision methods. For example, in Sofia Crespo's *Neural Zoo* series. These concepts are something that can also be observed in contemporary architecture work dealing with AI as a design method.

During the Summer of 2020, SPAN conducted a series of experiments interrogating the ability of GAN's to combine features of Baroque and modern plans. The thick, voluptuous pochés, pleats, and folds of the Baroque plan were successfully interpolated with the asymmetric features present in the modern plans.

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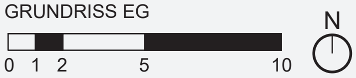


With this novel toolset comes a whole set of questions regarding its ontological and epistemological qualities. How do synthetic neural processes change the way architects conceive of their projects?

What does this mean? To unpack this problem, we need to divide it into its specific components. The assemblage of elements needs to be laid out—knolling the problem, so to speak. In neatly laying out the particular parts of the conversation, we hope to achieve a clear overview of the challenges that architecture is currently facing in the gestalt of a novel agent in design. Unpacking the pieces, unwrapping them, and cleanly positioning them on a flat surface allows for a clear overview. First, we need to discuss the ontological qualities of this problem. We are talking about architecture. To define architecture in itself borders on the naive, as it contains manifold problems reaching from purely practical to aspects that are cultural, aesthetical, and ephemeral. Instead, we will unpack the problem by closely looking into the meaning of neural networks and slowly and methodically working ourselves toward the aspect of the epistemology of the object we are observing and the elements of the paradigm and theory that they produce.

With or Without you - Dependency between Concrete and Abstract Objects

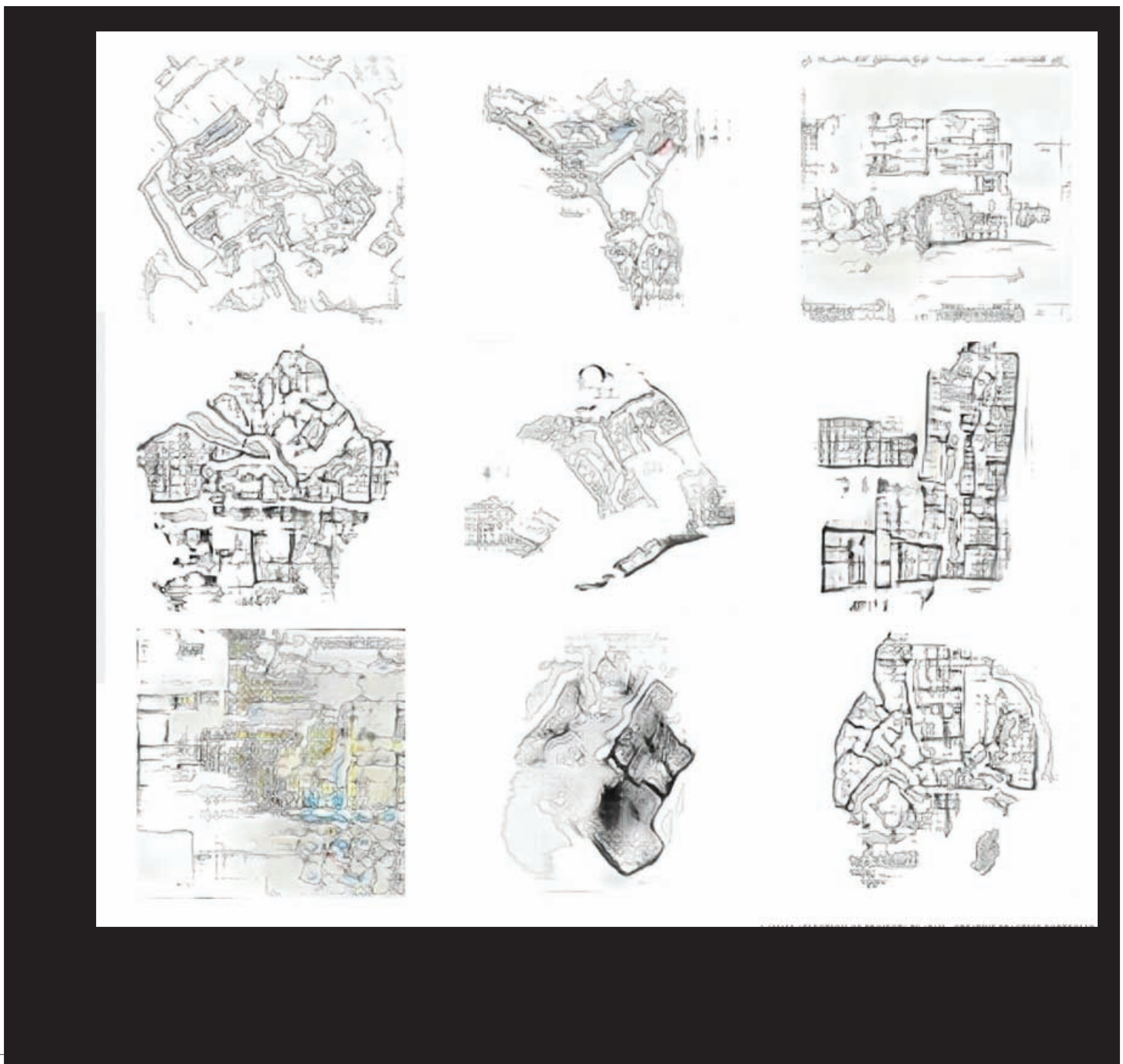
Neural Networks are Abstract Objects

As described above, NNs are prolific in recognizing relationships within a set of data. This is one of the reasons why this method is such a provocation for architecture. The traditional role of the architect is not very different. Architects are trained to recognize, understand, and process a large set of correlated information to assemble them in a project. Consider an architecture competition. When there is a call for competition, a set of rules in the form of programmatic, technical, social, and economic aspects is established. The specific program calls for the architect's knowledge, experience, and expertise to synthesize the rules in the form of a design idea. How does this knowledge emerge? Let's be upfront about the problem by assuming it is a specific competition. Maybe a theater? A very concrete object.[14] To create this concrete object, a set of abstract objects must first be established, such as plans and sections, which are based on calculations, numbers, and sets of diagrams demonstrating the spatial relationships. This produces a set of ontological dependencies, not only between concrete (material in the form of a building) and abstract (design, plans, sections) objects but also between the concrete object and the set of data present in the architect's mind. How many theaters did they design before; how many did they see; and how many images of theaters had been absorbed? Admittedly, this is a weak ontological dependency, but a dependency regardless. With this in mind, we can interrogate these entities—these examples of theaters in the architect's mind—by their properties.

Properties

What is meant by properties? Properties allow us to compare entities with each other. To stay with the example of the theater: The Burgtheater[15] in Vienna has an audience chamber and a stage tower. So do the Theater an der Wien, the Raimundtheater,[16] and the Volkstheater.[17] So we can assume that the audience chamber and the stage tower are properties of a theater[18] in that they are universally present in theaters. They can be shared by any number of theaters. To this extent, we can consider the theater as an object with a bundle of properties. (It would be easy to name more general properties of theaters, such as ticket boxes, lounges, rehearsal rooms, wardrobes, green rooms, administration, buffets, coat rooms, etc.) This might be true for every architecture that operates with a specific program, which ultimately represents a compresence relation responsible for the bundling. Of course, it is necessary to distinguish between the categorical and dispositional properties of an architectural object. When talking about categorical concerns, we discuss what something is like, meaning which kind of qualities it has. Dispositional qualities discuss the potentialities of an object (which, for what it is worth, can be tied to Manuel DeLandas conversation on *potentialities*). The Burgtheater has a precise shape, which is a categorical property, while its tendency to provoke thunderous applause (or at times booing) is a dispositional property. In regard to the discussion on AI and architecture, we are primarily interested in the categorical properties, as these can be examined by NNs

A series of examples resulting from a GAN learning features of modern and Baroque plans

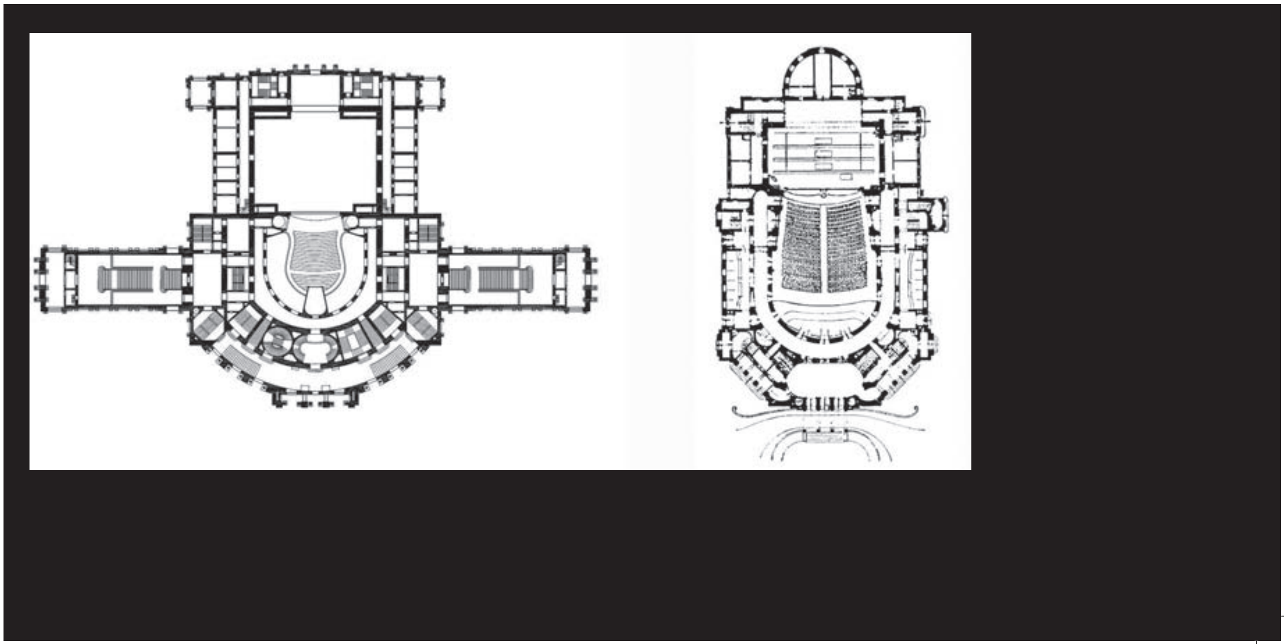


and can include shape, dimension, plan, section, umriss, pixels, etc. For this reason, we would also include color in the area of the categorical properties, as NNs divide color into their RGB values (numbers), which can be interrogated for their properties, thus positioning themselves within the conversation of abstract objects.

It's Complicated – about a Relationship

When talking about the properties of a piece of architecture, it seems evident that we should also discuss the relationships of these objects to each other. Traditionally in architecture, the discussion of the relationships between buildings and their environment, whether urban or rural, is described as the contextualization of a building. The Austrian Pritzker Prize winner Hans Hollein was a master in the minute and detailed interrogation of an architectural design in regards to the given context. I do not know how many hours I spent in Hans Hollein's office discussing the contextualization of a certain building. How does this axis meet that axis? How does the sun rotate around the building? What are the proportions and scales of the buildings around the project? As an intern in Hollein's office, I had to build dozens of physical models in blue foam with variations in how the building was contextualized in its surroundings.

Plan of the Burgtheater and the Raimundtheater in Vienna, Austria. Both have an audience chamber and a stage tower. We can assume that the audience chamber and the stage tower are properties of a theater in that they are universally present in theaters. They can be shared by any number of theaters. To that extent, we can consider the theater as an object with a bundle of properties.



Later I applied the same method in a digital context, allowing me to increase the possible variations into the hundreds—today it is datasets with thousands of examples. This increase in volume of variations demonstrates the transition from the age of data scarcity to the era of data abundance and the struggle to harness the inherent information. Or as Mario Carpo put it:

> The collection, transmission, and processing of data have been laborious and expensive operations since the beginning of civilization. Writing, print and other media technologies have made information more easily available, more reliable, and cheaper over time. Yet, until a few years ago, the culture and economics of data were strangled by a permanent, timeless, and apparently inevitable scarcity of supply: we always needed more data than we had. Today, for the first time, we seem to have more data than we need. So much so, that often we do not know what to do with them, and we struggle to come to terms with our unprecedented, unexpected, and almost miraculous data opulence.[20]

The response to Carpo's concern about how to harness the qualitative information of big data to inform the architectural project at hand is the use of neural networks. Not only do NNs process enormous amounts of data, but they also need enormous amounts of data to learn anything of value, whether it is relationships (contextualization), features (properties), or behavioral patterns. To this extent, the collision of the wealth of data produced by the architecture discipline throughout its history with processing methods borrowed from AI results in the field of neural architecture.

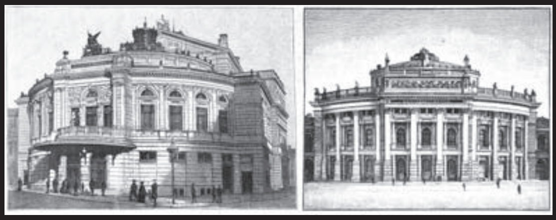
Coop Himmelb(l)au, series of models for the EZB Tower, Frankfurt, Germany. An example of versioning in a pre-computer era. These massing models were used as a design aid in regard to making decisions about massing, proportion, and significant features of a design.

About Wild Features and how to Capture Them

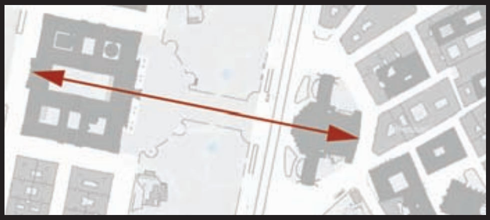
What does this mean for the relationships between these architectural objects? We can find a close connection here to the aspects of properties, in that both relationships and properties describe the character of the things they are applied to. Occasionally, properties are described as a special case of relations.[21] The example of the theater as the architectural object described in this chapter allows us to differentiate between internal and external relationships. The Burgtheater and the Raimundtheater are in an internal relationship of similarity to each other, as both are theaters.

The relationship is determined by the relata of similar features. As evident and simplistic as this insight seems to be, it is crucial for the operation and inner workings of neural networks. Features and their relationships to each other (for example, in an image) are what makes it possible to train a neural network to recognize a theater—or anything else it is trained to recognize, for that matter. Currently, this method is utilized in machine vision tasks that allow, for example, self-driving cars to recognize other cars, the street, signage, people, dogs, or cats—by understanding a family of internal relationships within a large number of images. In architecture, we can use this ability not only in an analytical sense but also in a creative sense, in that the flow of information can be reversed to *generate* a theater instead of *recognizing* one. I would like to emphasize the impact of such a motion, as it opens up questions about agency, authorship, aesthetics, and sensibility. Apart from the consideration of internal relationships, there are also aspects of external relationships. Particularly, this discussion would gravitate around the *contextualization* of a building. In contrast to internal relationships, external relationships in architecture are not defined by similarities. The Burgtheater is at the Ringstrasse; it stands in an axial relationship to the Rathaus but has no similarities to it. In fact, the architects of historicism, such as Gottfried Semper (Burgtheater) and Friedrich von Schmidt (Rathaus), specifically used very different features to evoke the meaning of the content of the building. While the Burgtheater was designed in a Neo-Renaissance style (because, Culture, you know) the Rathaus (city hall) was designed in a Neo-Gothic style, evoking the origin of citizen representation in medieval Flemish towns (what this has to do with Vienna is up to a Friedrich von Schmidt scholar to explain).

Burgtheater and Raimundtheater—similar but not the same.



Relationship of Burgtheater and Rathaus, Vienna, Austria.



Things, Facts, and the Ontology of Neural Networks

Now, since it has been established that NNs are abstract objects defined by their properties and their relationships, one more question remains: Is neural architecture a thing or a fact? Why is this relevant to the conversation laid out in this book? It is very alluring to declare that every architecture is a thing, and thus every aspect and part of it is a thing, too. But is this really so? What about plans, calculations, proportions, simulations? Are those things or facts? Well, maybe it would help first to briefly define what is meant by things and facts. Thing ontologies discuss the possibility that the universe is made up of a plurality of discrete objects.[22] To this extent, the paradigm example of a thing ontology would be atomism.[23] This sits in contrast to non-thing ontologies, which consider that an object is not necessarily an assembly of objects, but could also be one continuous object. For our purposes, let us stick to the idea of NNs being things, in that they represent an assembly of objects. These discrete objects can be the databases of individual images or individual *obj*[24] models (as in the research on GraphCNN design methods[25]) or the numerical data of images (in the form of their RGB values) or the layers in the networks that fulfill different tasks, such as edge recognition. How are these things and not facts? Or are they both? If we take a materialist point of view (and materialist is interchangeable with realist in this conversation), I can posit that the calculation of NNs is based on material processes: electrically charged matter moves, creating a set of physical phenomena resulting in an electric current, which, in turn, is used to run computers and their respective GPUs, which are made of silicon, tantalum, and palladium, and calculate the numbers laid out by a set of algorithms known as neural networks.

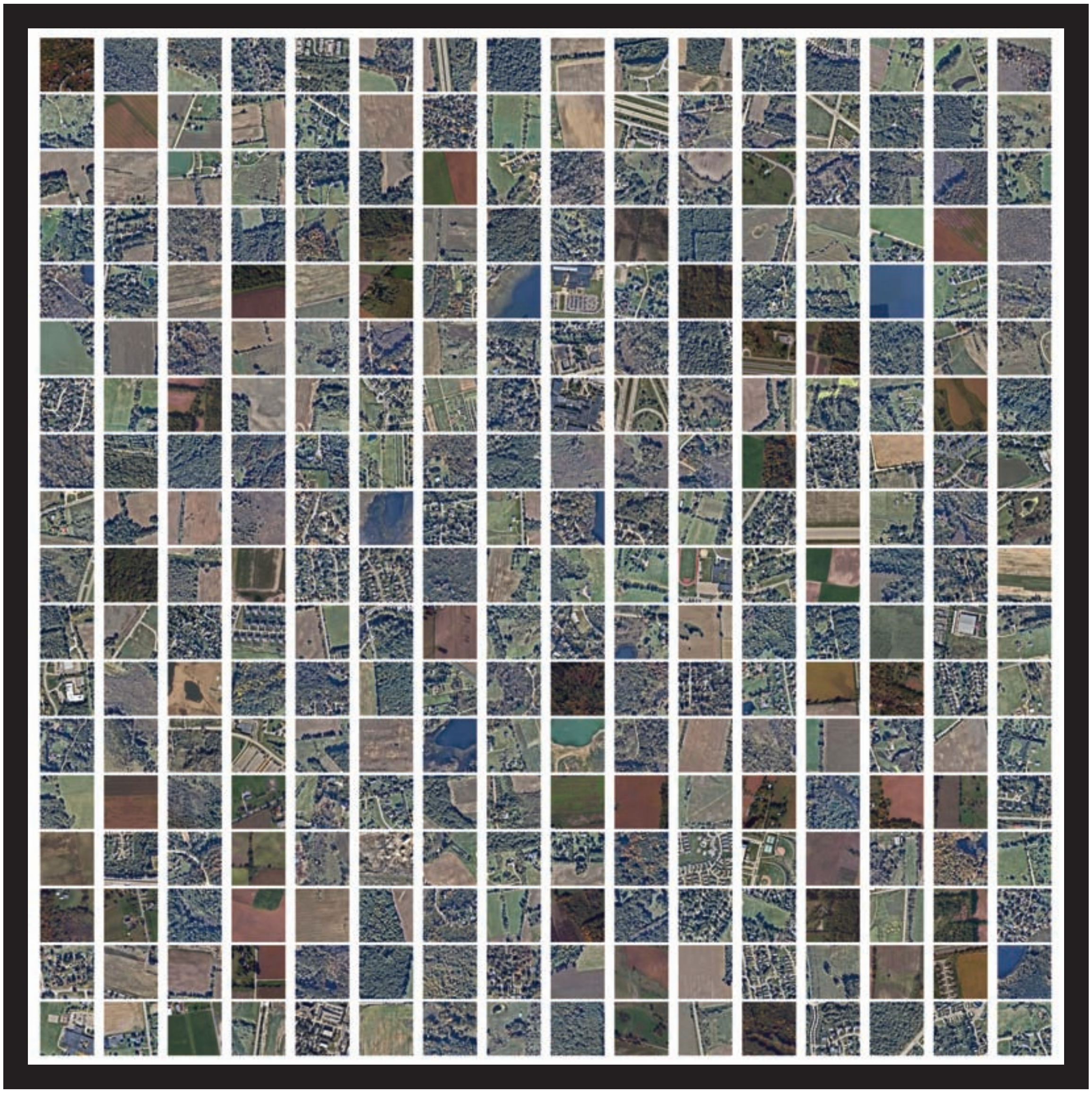
Daniel Bolojan, *Semantic Feature Learning*.

Immanuel Koh, *3DGAN chair*.

All of these are things operating in discrete processes to create a result. The result can be the successful recognition of a car, a person, a dog, a sign or a voice—or the creation of an image that we describe as art (see Sofia Crespo and Mario Klingemann), music (see Dadabots, YACHT, and Holly Herndon) or architecture (see SPAN, Daniel Bolojan, Immanuel Koh, etc.). What if neural networks are not things but facts? What is the difference? Things are generally put in contrast with the properties and relations they instantiate,[26] in contrast to facts, which are described as adhering to things in combination with their properties and relations as their constituents. In short, facts swallow things and their properties/relations versus things in themselves maintain independency in a flat ontology.

What about Aesthetics, Agency, and Authorship?

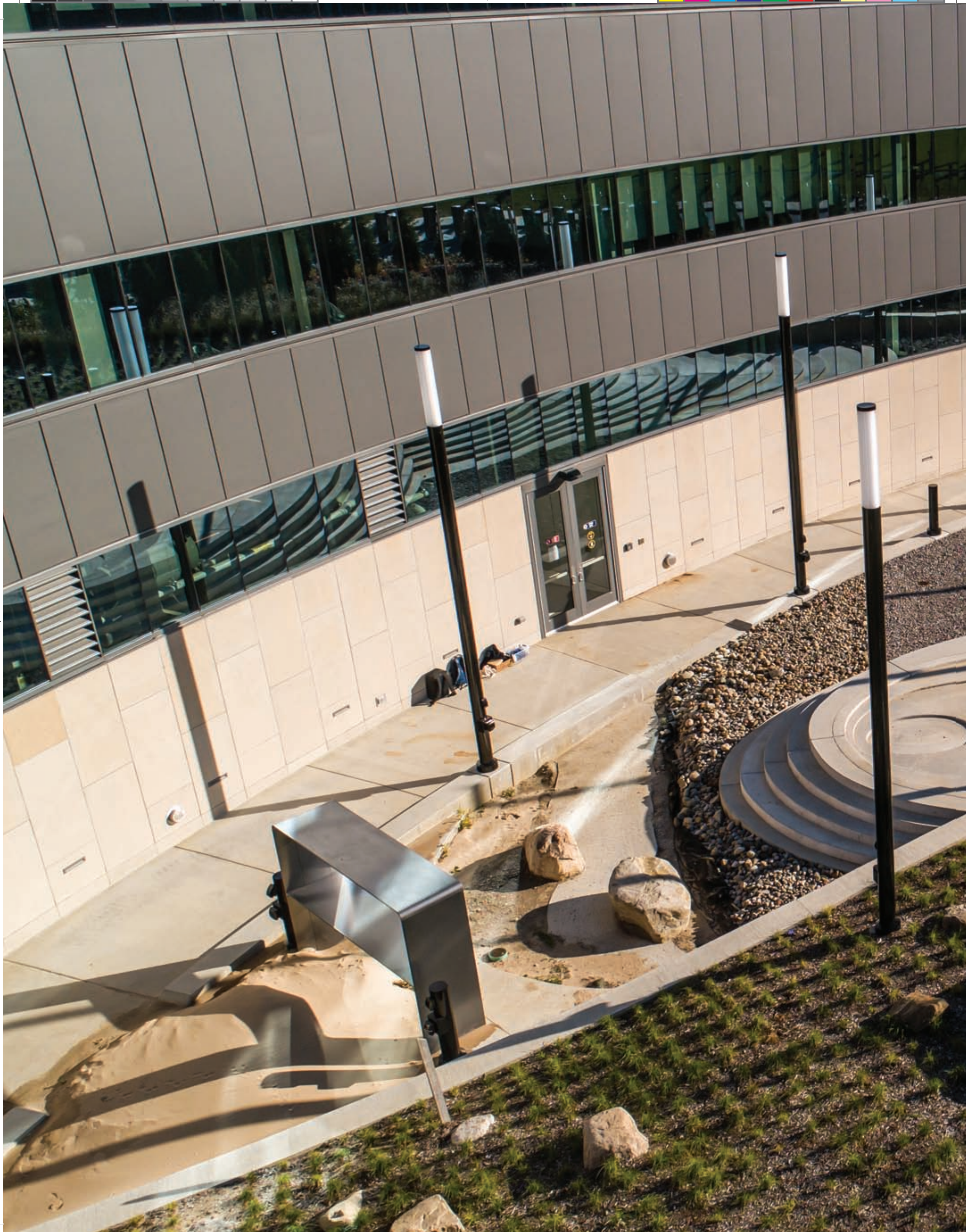
For the conversation on neural networks and their meaning for the architecture discourse, there are three key takeaways here: NNs are things—they are abstract objects with properties; they maintain internal and external relationships within a flat ontology; and they constitute discrete assemblies.[27] Now, after interrogating the ontological frame in the area of neural architecture, it might be time to ask if we actually gathered any knowledge from the frame we laid out. Do we know more now that we have explored concepts such as objects, properties, and relations between elements at play in neural architecture? Most importantly, the previous interrogation already hinted at some of the epistemological questions that arise out of the work with neural networks, such as the aesthetics, agency, and authorship of the resulting project. Relevant to the frame of conversation in this essay are the positions of Roland Barthes and Michel Foucault in regard to the nature of authorship and the author at large. These critics interrogated the role and relevance of authorship pertaining to the interpretation or meaning of text; this can be expanded to all areas of artistic production including, for example, architecture in the present text of this book. Barthes for example attributed meaning to the language and not the author of the text. Instead of relying on the legal authority to exude authorship,[28] Barthes assigns authority to the words and language itself. Foucault's critical position versus the author can be found in his essay "What is an Author."[29] Foucault argues that all authors are writers, but not all writers are authors, echoing a broadly discussed sentiment in architecture: ***not every architecture is a building, and not every building is architecture.***



SPAN, the *Robot Garden.* A grid of satellite images of various topographic conditions that were used to train a neural network to dream features onto the site. The aesthetics of the image grid is a fundamental element of current work on AI and architecture. Partly the result of the, at times, tiny resolution of images used in ML (256x256px) and partly a continuous decision to find a method to represent big data and large datasets.

Aesthetics of Neural Architecture

39 The term aesthetics is a highly charged one in the architecture discourse. This includes considerations such as the antithetical position of Peter Eisenman, who demanded that figures of architecture should be read rhetorically instead of aesthetically or metaphorically,[30] and Anthony Vidler's discussion of aesthetics along the lines of its functional and metaphysical properties, such as the sublime.[31] The authors would first propose to interrogate the etymology of the word aesthetics in order to unlock its meaning. Greek in its origin (aisthēsis), the term aesthetics literally means "perception by the senses," which, by the way, brings us right back into the discussion of feature recognition. The German philosopher and father of our modern understanding of the term aesthetics as the science of sensory knowledge directed toward beauty, Alexander Gottlieb Baumgarten, considered art to be the "perfection of sensory awareness." [32] It is important to understand that the term aesthetics has undergone a series of mutations since Baumgarten's time and that even contemporaries of Baumgarten, such as Immanuel Kant, had a very critical position toward Baumgarten's definition as it lacked—according to Kant—any objective rules, laws, or any principles for that matter, able to describe natural or artistic beauty.[33] In contrast to the objective usage of the term aesthetics, Kant relied on its use in a subjective manner as it relates to the internal feeling of pleasure or displeasure and not to any qualities in an external object. The assertion of Baumgarten that the three guiding principles of aesthetics are "good, truth, and beauty" appears essentially naive in the contemporary age.[34] Thus, it is surprising to me to find this branch of conversation on aesthetics still being perpetuated by critics such as Roger Scruton[35] and Yael Reisner.[36] In general, it could be stated that the meaning of aesthetic has become equivocal to connoting something whose appearance displays a particular set of qualities that evoke a response in the observer. The cognitive activity of aesthetical observations is a priori an exercise that is both perceptual as well as emotional. Considering this, it seems evident how this activity is profoundly connected to epistemological considerations. Let us return to the example of theater in Vienna. The highly critical theater audience in Vienna "knows" a thing or two about theater and is not shy about classifying the play as "good" or "bad" (at times it is not so much about the content of the play, but rather the quality of the performance). Theater critics are highly respected. Emotions can run high at the end of a premiere, with a mixture of applause and booing. (After the premiere of Thomas Bernhard's Heldenplatz at the Burgtheater, I think I remember fists flying.). In general, the belief that art can change the perception of the world is widely recognized, up to the point that good art can engender beliefs about the world, and thus can provide knowledge about the world. The aesthetics of neural architecture are particularly interesting in allowing us to reflect on aspects of human and artificial ingenuity, the position of architecture after several millennia of history, and its further development in a posthuman era. It is an aesthetic that oscillates between the familiar and moments of estrangement in an architecture that is at the same time alien but full of recognizable features.



When I talk about the term sensibility, it is specifically geared toward aspects of artistic sensibility. In the previous section, I outlined a historical and theoretical development along the lines of Baumgarten's and Kant's definitions of aesthetics, which concluded in the insight that aesthetics is, at its very core, a theory of sensibility—evoking a response in the observer. This assertion discusses the arts of the past as much as the arts of the present, recognizing aesthetic value as a specific feature of all experience. Or as Arnold Berleant put it,

> Such a generalized aesthetic enables us to recognize the presence of a pervasive aesthetic aspect in every experience, whether uplifting or demeaning, exalting or brutal. It makes the constant expansion of the range of architectural and of aesthetic experience both plausible and comprehensible.[37]

How is it possible to scrutinize the term sensibility within this frame of thinking? What I mean by sensibility is perceptual awareness developed and guided through training and exercise. To this extent, it is certainly more than simple sensual perception, and closer to something like a guided sensation—an education that has to be continuously fostered, polished, and extended through encounters and activities, in order to maintain the ability to execute tasks with an aesthetic sensibility. In Western traditions, this ability is attributed primarily to the arts—to painting, sculpture, music, literature, and so on—with architecture being this strange animal living somewhere between engineering and the arts. We can interrogate these territories of human production in regard to their fashions, styles, etiquettes, and changing behavioral patterns as a result of transforming sensibilities, giving rise to novel movements and entire epochs in the arts in general and architecture in particular.

◂ Previous page: SPAN, the *Robot Garden.* The basis for this project is a series of experiments with neural networks conducted between 2018 and 2019. Exploring the possible options in feature recognition, deep dreaming, style transfer, and 2D–3D transformations. This project was crucial in developing methods to design architecture elements using AI techniques.

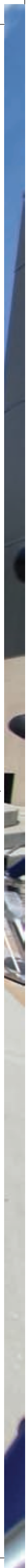
Agency in Neural Architecture

In Western philosophical tradition, causal chains do not produce our choices, as would be the case with objects responding to natural forces, thus giving us agency. Free will and agency are closely related, but not identical. They share traits, like cousins of the same family, in that agency is undetermined and significantly free. In contrast to inanimate objects, humans can make decisions and enforce them in the world. For the moment, we will leave out the metaphysical question as to how humans make decisions. In any case, it entails moments of moral agency, as particular acts of human agency require thought and consideration about the outcomes. For this paper, we rely on agency as part of the debate on action theory.[38] This can be exemplified by the philosophical traditions established by Georg Wilhelm Friedrich Hegel[39] and Karl Marx[40] that consider human agency in a collective fashion; thus, for our frame of thinking, we consider the relationship between humans and neural networks as such a collective with a particular agency. In particular, we focus on the argument pertaining to the ideas of materialism and realism, which we will unpack in more detail.

Neural Architecture is a New Paradigm

New paradigms have the habit of emerging when the existing paradigm has run its course, which begs the question: What is the current paradigm? But let's focus on what neural architecture brings to the table regarding a substantial innovation in the discipline of architecture. For one, it critically interrogates the role of the architect in the creative process of architecture design. Neural architecture embraces the possibility of design methods that are deeply informed by existing information in the form of databases and understands that the artificial modeling of neural processes can aid in harnessing the information in big data.

Interestingly, the results do not resemble historic examples, and thus the methodology is not a repetition of postmodern tropes, such as collage, quotes, and ironic assemblies. The results rather construct a frame around aspects of **defamiliarization and estrangement**, in that we are able to recognize certain features without it being a copy. We are still at the very beginning of this new paradigm of architectural production; the first built examples are currently emerging.[41] Many specific problems need to be solved, such as the relationship between interior and exterior, as neural networks based on images are only able to solve one of the two at a time or the problem of utilizing 3D models in this paradigm. Solutions are under development as these lines are written, and I look forward to watching how this new paradigm will impact the built environment in the upcoming years.



◂ SPAN, the *Robot Garden*. A testing ground for robots at Michigan Robotics' new Ford Research building. This project served as a proof of concept project for the theoretical ideas that form the basis for neural architecture—harnessing the power of big data and channeling them through Artificial Neural Networks to design architecture elements.

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